Wave Optics

FYS2150

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Abstract

Wave optics, also known as physical optics, is a fundamental branch of physics that deals with the nature and properties of light . In this set of experiments , we will be focusing on the behavior of laser light when passes through different types of slits and gratings, the observation of emission lines of Hydrogen, Helium, and the application of Fabry-Pérot (FP) interferometry to determine the Bohr magneton μ_B of a Cadmium emission lines after placing it in a magnetic field .

1 Introduction

The first part of our investigation explores the diffraction and interference patterns produced by laser light as it passes through different types of slits and gratings(singular,parallel and circular).This phenomenon, known as the diffraction of light. By varying the characteristics of the slits and gratings, we can observe and analyze the changes in the resulting diffraction patterns.

Next, we turn our attention to the emission spectra of Hydrogen and Helium. When these elements are excited, they emit light at specific wavelengths, creating unique emission lines that serve as a "fingerprint" for each element. By passing this emitted light through a diffraction grating, we can separate it into its component wavelengths and observe these emission lines. This part of the experiment provides a practical demonstration of how these principles are used in spectroscopy to identify elements.

Finally, we will explore Zeeman-effect by employ Fabry-Pérot interferometry, a powerful technique in wave optics, to determine the Bohr magneton μ_B . The Bohr magneton is a physical constant and a key quantity in quantum mechanics, representing the fundamental unit of magnetic moment for an electron. By using FP interferometry, we can measure this quantity with high precision, demonstrating the practical applications of wave optics in the realm of quantum physics.

2 Theory

The wave theory of light, as proposed by Huygens and later developed by Fresnel and Young, provides the theoretical basis for the phenomena of interference and diffraction. This theory treats light as a wave that can interfere constructively or destructively with other light waves.

2.1 Interference , Diffraction

Interference is the phenomenon that occurs when waves pass through slits. According to Huygens' principle, any point on a wavefront can be considered as the source of secondary wave that spreads out in all directions. The principle of superposition states: When two waves are in phase (i.e., two peaks or two troughs align), they interfere constructively, resulting in a wave with a greater amplitude. When they are out of phase (i.e., the peak of one wave aligns with the trough of the other), they interfere destructively, cancelling each other out and resulting in a wave with a reduced amplitude.

The interference pattern can be detected on a screen and is described by the intensity function:

$$E_N(x) = E_1(x)F_N(x) \tag{1}$$

where:

- N is the number of slits,
- $E_1(x)$ is the diffraction curve.
- $F_N(x)$ is the interference curve.
- x is the distance on the x-axis.

2.2 Diffraction from one slit:

For one slit($N = 1$),the intensity is given by:

$$E_N(x) =\propto (\frac{\sin(acx)}{acx})^2 \tag{2}$$

where:

- $c = \frac{\pi}{\lambda R}$
- R is the distance between the slit and the screen.

- λ is the light wavelength.
- x is the distance on the x-axis.
- a is the width of the slit.

This function has a minima(dark stripe) at :

$$x = \frac{n\lambda R}{a}, where n = \pm 1, \pm 2.. \tag{3}$$

2.3 Diffraction and Interference from to two or more slits:

When light passes through two or more parallel slits, each separated by a distance$A$ and all having the same width a, each slit acts as a source of a new wave. The superposition of these waves forms an interference pattern. Constructive interference occurs when the waves are in phase, resulting in a pattern of maximum . The intensity function is given by

$$E_N(x) = E_1(x)F_N(x) \propto (\frac{\sin(acx)}{acx})^2(\frac{\sin(NAcx)}{\sin(Acx)})^2 \tag{4}$$

This function reaches a major maximum when : $x = \frac{m\lambda R}{A}$,where $m = 0, \pm 1, \pm 2...$

Between two major maxima , we have $(N-1)$ zero points and $(N-2)$secondary maxima .The secondary maxima diminish as N increases .

2.4 Diffraction from circular aperture:

When light passes through a circular aperture, it creates a diffraction pattern consisting of a central maximum surrounded by concentric dark and bright rings as shown in Figure below. The intensity of the light at a point in this pattern depends on the angle from the central maximum, the wavelength of the light, and the size of the aperture.

$$H(\omega) \propto [\frac{j_1(\omega)}{\omega}]^2, \omega = \frac{nar}{\lambda R} \tag{5}$$

- j_1 is the Bassel function of the first order.
- r is the distance from the center of the diffraction pattern in the plane of observation.

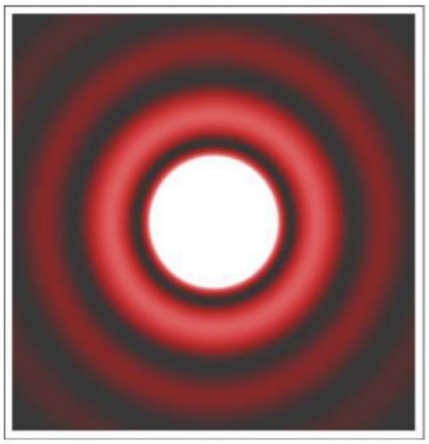


Figure 1: Diffraction pattern from a circular slit

Zero points of this function occur when $\omega_0 = 3.832,7.016,...$These points correspond to dark rings on the figure above. The formula for the radius of the m-th dark ring in a diffraction pattern due to a circular aperture is given by:

$$r_m = 1.22 \cdot m \cdot \lambda \cdot \frac{R}{d}$$

where:

- r_m is the radius of the m-th dark ring,
- m is the order of the dark ring ($m = 1$ for the first dark ring, $m = 2$ for the second, etc.),
- λ is the wavelength of the light,
- d is the diameter of the circular aperture,
- R is the distance from the aperture to the screen.

screen. To get the diameter of the dark ring, you simply multiply the radius by 2:

$$d_m = 2 \cdot r_m = 2.44 \cdot m \cdot \lambda \cdot \frac{L}{D} \tag{6}$$

2.5 Balmer lines

The Balmer lines are a series of spectral emission lines created by atom when an electron descends to the second energy level ($n = 2$) from higher energy levels. These lines fall in the visible spectrum. The wavelengths of the characteristic radiation from hydrogen can be found by applying The Rydenberg formula:

$$\lambda_n = \frac{1}{R[(\frac{1}{2})^2 - (\frac{1}{n})^2]} \tag{7}$$

- $R = 1,097.10^7 m^{-1}$, is Rydenberg constant.
- $n = 3, 4, 5, ...$ Is the principle quantum number of the initial state and it is always greater than 2.

2.6 The Zeeman Effect

The Zeeman effect is a magneto-optical phenomenon which we can explain using quantum mechanics theory. In short, the effect is about that the energy associated with a "jump" from a electron shell of another can be affected by an external magnetic field.

We study the case where the magnetic field is normal on the light direction. The light is then plane polarized, and that the electric field of the two new lines is normally on the electric field of the light of wavelengthλ_0 . Using a polarization filter, we filter out the line of wavelength λ_0 and consider in it following only the remaining lines of frequencies $v_a$and v_b. We can find the frequency difference:

$$\Delta v = v_a - v_b = \frac{2\mu_B B}{h} = \frac{c}{2t}\frac{d_2^2 - d_1^2}{d_3^2 - d_1^2} \tag{8}$$

We define $\delta = \frac{d_2^2 - d_1^2}{d_3^2 - d_1^2}$, $\Delta v = \frac{c}{2t}\delta$,$\mu_B = \frac{h\Delta v}{2B}$, Here $t = 2.9 \pm 0.1mm$ is the distance between the two plates in the FP-interferometer.

3 Experimental

3.1 Diffraction. Single slit of different widths

We start by setting up our experiment as shown in the figure 1 below.

In this expirement , we use :(THORLABS, HRR120, HeNe laser) . with a chosen wavelength $\lambda = 632,8nm$. Between the source and the screen , we place a (PASCO Precision Diffraction Slits) with four distinct single slits

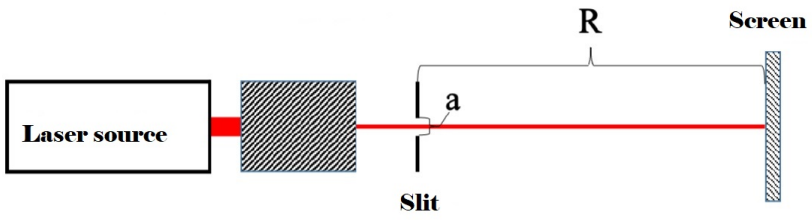


Figure 2: Diffraction from a Single slit

of varying widths ($a = 0.02, 0.04, 0.08, 0.16mm$) such that the light passes through one slit at a time. The apparatus can be rotated to switch between the four different slits. We measure the distance R between the source and the screen with a laser pointer yielding a result of :

$$R = 1.7m \pm 0.5mm$$

Subsequently, we measure the distance between two dark regions (two minima) for each slit by placing a white paper where the light is projected . Using these measurements, we calculate the corresponding slit width. Finally, we compare these calculated values with the theoretical values provided for each slit.

3.2 Diffraction,interference from two parallel slits

In this experiment , we use the same devices as previously but use four different combinations of double slits with a distance between slits$A$ and the slit width a:

- $A = 0.25mm, a = 0.04mm.$
- $A = 0.5mm, a = 0.04mm.$
- $A = 0.25mm, a = 0.08mm.$
- $A = 0.5mm, a = 0.08mm.$

Once again, we rotate the apparatus to switch between the four different slit combinations. For each set, we count the number of maxima generated and compare these counts with the theoretical numbers.

3.3 Diffraction from a circular slit

In this part of the experiment, we use a different apparatus to investigate the diffraction from a circular slit of two different diameters $d = 0.2mm, d = 0.4mm$ We measure the diameter of the dark regions and compare these measurements with the given theoretical values.

3.4 Balmer lines

The grating spectrometer, primarily consists of a collimator, a grating or prism, and binoculars equipped with a reticle in the eyepiece. When a grating is positioned perpendicular to the axis of the collimator, it deflects the spectral components of the light emitted from the lamp in different directions. By adjusting the binoculars to "infinity", we can accurately read the angles corresponding to these directions on the spectrometer's degree scale. This setup allows for the precise measurement and analysis of the spectral components of light, including the Balmer lines. When the light form the lamp passes through the grating, The diffraction grating diffracts the incident light into identical spectra on either side of the line of the undiffracted beam. By rotating the binoculars to both sides, we can observe the spectral lines. These observed lines can then be used to calculate the wavelength of the light.

The same procedure was applied to both Hydrogen and Helium .

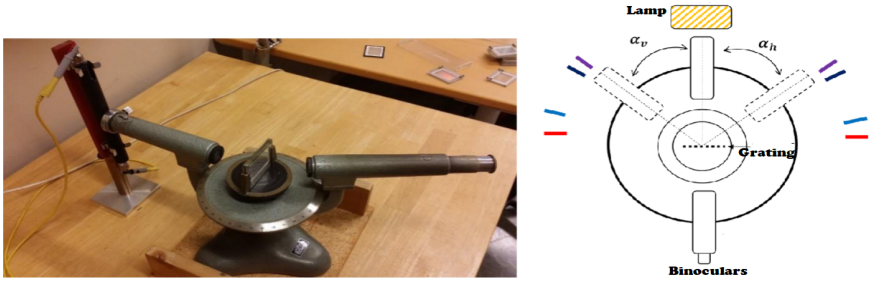


Figure 3: Spectrometer used to measure angle of diffraction

3.5 The Zeeman effect

In our experiment, we utilized a Fabry-Pérot (FP) interferometer, as shown in Figure 5. The light source for this setup was a cadmium lamp with wavelength$\lambda = 643.8nm$. The emitted light from this lamp was first passed

through a focusing lens, which used to concentrate the light beam. Following this, the light was filtered based on color and polarization to ensure that only the desired light characteristics were allowed to pass through. The filtered light was then diffracted and subsequently focused through another lens.

The resulting light was captured by a sensitive camera, which was connected to a computer for data recording and analysis. To enhance the visibility of the diffraction lines, we adjusted various settings such as exposure and contrast. This careful calibration allowed us to obtain clear and distinct diffraction lines, thereby facilitating a more accurate analysis of the light's properties.

In addition to the setup, an electromagnet, connected to a power source, was positioned around the light source. The calibration curve for this electromagnet is depicted in Figure (4). By subjecting the cadmium lamp to a magnetic field, we were able to clearly observe the phenomenon of Zeeman splitting. This was achieved by adjusting the polarization filter in such a way that the original diffraction patterns became invisible, while the newly formed "split" lines were fully illuminated. This adjustment allowed us to clearly observe and analyze the effects of the magnetic field on the light emitted by the cadmium lamp. Two pictures were taken for two different currents : $I = 3A, I = 4A$ these two pictures will be analyzed and used to determine the inner and outer diameter.

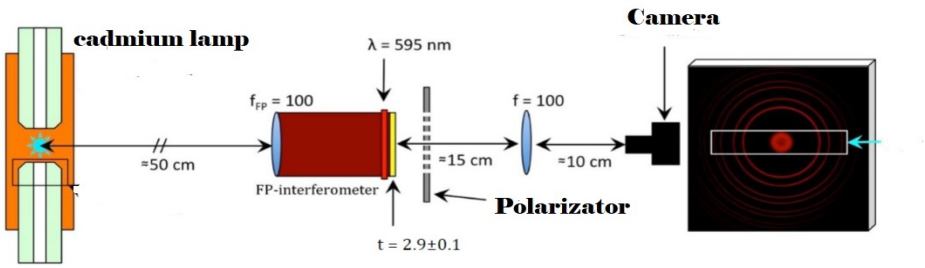


Figure 4: Fabry-Pérot FP-interferometer

4 Results

4.1 Diffraction. Single slit of different widths

The distance between the laser source and the screen was measured to be$R = 1.7m \pm 0.5mm$. By measuring the width of the distance on the

screen and then divided by the number of minima for each slit , we obtained the following results:

$$[5.5, 2.64, 1.364, 0.678]cm \pm 0.1cm$$

by applying equation(3) and solving for$a$, we obtained results shown in Table(1) below. $(\frac{\Delta a}{a})^2 = (\frac{\Delta R}{R})^2 + (\frac{\Delta x}{x})^2 \implies \Delta a = |a|\sqrt{(\frac{\Delta R}{R})^2 + (\frac{\Delta x}{x})^2} = 3.45 \times 10^{-4}mm$

Table 1: Single slit

x [cm]	Calculated Width [mm]	Theoretical Width [mm]
5.5	$0.019 \pm 3.45 \times 10^{-4}$	0.02
2.64	$0.04 \pm 3.45 \times 10^{-4}$	0.04
1.364	$0.078 \pm 3.45 \times 10^{-4}$	0.08
0.678	$0.158 \pm 3.45 \times 10^{-4}$	0.16

4.2 Diffraction,interference from two parallel slits

Using the combinations ;

- 1,$A = 0.25mm, a = 0.04mm.$
- 2,$A = 0.5mm, a = 0.04mm.$
- 3,$A = 0.25mm, a = 0.08mm.$
- 4,$A = 0.5mm, a = 0.08mm.$

We obtained results will be shown in the table(2) below:

Table 2: Two parallel slits

Combination	Observed No. Peaks	Expected No. Peaks
1	13	11
2	25	23
3	5	5
4	11	11

4.3 Diffraction from a circular slit

We measured the distance between the laser source and the screen with a laser pointer to be

$$R = 1.83m \pm 0.5mm$$

We used in this experiment two slits: $d_1 = 0.2mm$ and $d_2 = 0.4mm$. We obtained the calculated diameters by applying equation(6).The results are shown in Table(3) below

Table 3: Circular Slit

slit diameter [mm]	d_1[cm] $\pm 0.1mm$	d_2[cm]$\pm 0.1mm$	d_3[cm]$\pm 0.1mm$	d_4[cm]$\pm 0.1mm$
0.2	1.31	2.51	3.76	4.77
0.4	0.75	1.225	1.83	2.38
slit diameter [mm]	d_1Calculated[cm]	d_2Calculated[cm]	d_3Calculated[cm]	d_4Calculated[cm]
0.2	1.42	2.82	4.23	5.64
0.4	0.70	1.412	2.12	2.82

4.4 Balmer lines

1- Hydrogen lines : In this experiment , we used $\theta_0 = 193.5°$ Where a_v,a_h left angular position and right angular position of emission lines respectively,θ corresponds to the difference between the two angular positions. We measure :

- Green,$a_v = 158.3 \pm 0.1°$,$a_h = 228.5 \pm 0.1°$.
- Red,$a_v = 142.3°$,$a_h = 243°$.
- Violet $a_v = 162.7 \pm 0.1°$,$a_h = 224.4 \pm 0.1°$.

Using the formula :

$$\theta = \frac{a_v - a_h}{2}$$

$$\lambda = \frac{d \sin\theta}{m}$$

, where $m = 1$,The grating constant $d = 8.46 \times 10^{-7}$.

$$\Delta\lambda = \lambda \frac{\Delta\theta}{\tan\theta}$$

We use a Python script to calculate the correspondingλ, we obtained :

- Red ,$\lambda = 652 \pm 0.002nm$.
- Green ,$\lambda = 487 \pm 0.002nm$.
- Violet,$\lambda = 434 \pm 0.002nm$.

2-Helium lines : We used $\theta_0 = 223.4°$. Where a_v,a_h left angular position and right angular position of emission lines respectively,θ corresponds to the difference between the two angular positions. We obtained the following results as shown in table(4) below.

Table 4: Helium lines

color	$a_v \pm 0.1°$	$a_h \pm 0.1°$	$\lambda \pm 0.002[nm]$
Blue-violet	171.4	234.4	439±0.002
Blue	170.7	234.5	448 ±0.002
Blue-Green	168.6	236.4	442±0.002
Dark-Green	166.7	238.0	494±0.002
Light-Green	165.8	238.7	500±0.002
Yellow	157.6	245.7	589±0.002
Red	147.7	253.0	673±0.002

4.5 The Zeeman Effect

The pictures from Fabry-Perot (FP) interferometer are shown in Figur(5) below:

Table 5: Zeeman results

I [A]	$B(I)$ [mT]	$d_1(I)$ [px]	$d_2(I)$ [px]	$d_3(I)$ [px]	μ_B [J T^{-1}]
3	533	394 ± 1	470 ± 1	621 ± 1	$9.162 \pm 0.14 \cdot 10^{-24}$
4	694	332.0 ± 1	475.0 ± 1	613.5 ± 1	$1.06 \pm 0.18 \cdot 10^{-21}$

By using the formula

$$\Delta v = \frac{c\delta}{2t}$$

, with $t = 2.9 \pm 0.1mm$ is the distance between plates in FP-interferometer. We obtained: $\Delta v_1 = 0.285$, $\Delta v_2 = 0.433$ Finding the uncertainty is quit challenging.

$$\mu_B = \frac{h \cdot \Delta v}{2B} \tag{9}$$

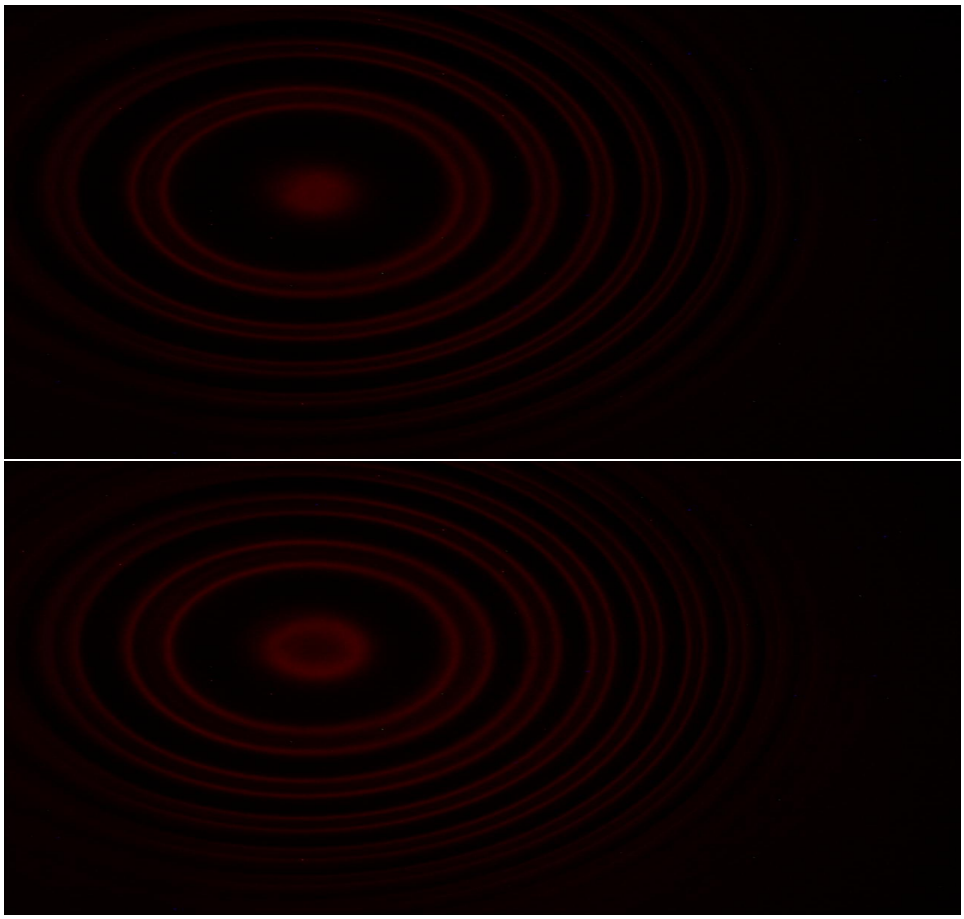


Figure 5: Picture from (FP) interferometer shows diffraction lines due to σ-transitions for various magnetic fields

Assuming that the uncertainties in h, Δv, and B are δh, $\delta(\Delta v)$, and δB respectively, the uncertainty in μ_B can be calculated using the general formula for the propagation of uncertainty:

$$\delta f = \sqrt{\left(\frac{\partial f}{\partial x}\delta x\right)^2 + \left(\frac{\partial f}{\partial y}\delta y\right)^2 + \left(\frac{\partial f}{\partial z}\delta z\right)^2 + \cdots} \tag{10}$$

Applying this to μ_B, we get:

$$\delta\mu_B = \sqrt{\left(\frac{\partial \mu_B}{\partial h}\delta h\right)^2 + \left(\frac{\partial \mu_B}{\partial \Delta v}\delta(\Delta v)\right)^2 + \left(\frac{\partial \mu_B}{\partial B}\delta B\right)^2} \tag{11}$$

$$\delta\mu_B = \sqrt{\left(\frac{\Delta v}{2B}\delta h\right)^2 + \left(\frac{h}{2B}\delta(\Delta v)\right)^2 + \left(-\frac{h\cdot\Delta v}{2B^2}\delta B\right)^2} \tag{12}$$

5 Discussion

5.1 Diffraction from a single slit

From Table(1), we can see that the measured slit widths, which are based on the diffraction pattern observed on a screen, closely align with the theoretical values. There is a good agreement between the theoretical and measured values for the slit width of 0.04 mm, but there is a larger deviation for the slits with widths of 0.08 and 0.16 mm.

This discrepancy can be attributed to several factors. The method of measurement, which involves placing a white paper on the screen, is prone to errors. For instance, the paper may not have been placed perfectly flat against the screen. Furthermore, the ruler used as a measuring tool may not have been accurate or sensitive enough to measure lengths in the millimeter range.

In addition, the lines of the diffraction pattern may not have been sharp enough to enable precise measurements. These factors, among potentially others, could have contributed to the observed deviation between the theoretical and measured slit widths.

5.2 Diffraction from two slits

From data in Table 2, the measured peaks for combinations 3 and 4 align perfectly. However, there is a noticeable deviation from the expected theoretical number for combinations 1 and 2, which have a slit width of 0.04mm. Instead of the anticipated number of peaks, two more peaks were observed.

5.3 Diffraction from circular slit

The process of measuring the width of dark rings, denoted as $d1$,$d2$,$d3$ and $d4$ yielded values of [1.31,2.51,3.76,4.77]cm respectively for the slit $d = 0.2mm$ when calculations yielded[1.42,2.82,4.23,5.64] . For the slit $d = 0.4mm$ the measurement yielded :[0.75,1.225,3.76,4.77]cm,respectively while calculated values were: [0.7,1.412,2.2,2.82]. It is easy to observe the deviation between calculated and measured values. One possible explanation for this discrepancy could be the distance between the projecting screen and the slits.The screen was not positioned far enough from the slits, it could have affected the precision of the measurements, making it difficult to accurately measure the diameter of the diffraction rings. This could have led to the observed deviation in the measured diameters from their theoretical values. Another explanation could be the paper placed on the screen was not perfectly flat.That could have affected the accuracy of our measurement.

5.4 Balmer lines

We made measurements for the wavelengths of the Hydrogen emission lines, obtaining the following results

$$Red, \lambda = 6520.002nm$$

$$Green, \lambda = 4870.002nm$$

$$Violet, \lambda = 4340.002nm$$

These measurements were then compared to their respective theoretical values: 656.34nm for Red, 486.17nm for Green, and 434.21nm for Violet. Upon comparison, it was observed that our measurements align relatively well with the theoretical values, with only minor deviations. These slight discrepancies could be related to a couple of factors. One potential source of error could be the accuracy of the measuring instrument used. Additionally, human error, especially the limitations of the human eye in making precise observations, could also contribute to these uncertainties. For Helium spectrum,the violet line was not observable.

5.5 The Zeeman Effect

We carried out a series of measurements to calculate the Bohr magneton μ_B. These measurements partially align with the theoretical value provided by CODATA, which is:

$$\mu_B = 9.274009994 \cdot 10^{-24} j/T$$

Our measured results: For $I = 3A$ we measured $\mu_B = 9.162 \cdot 10^{-24}$ which nicely close to the CODATA value. For $I = 4A$, we got $\mu_B = 1.069 \cdot 10^{-}23$ which is a huge deviation from the real value.

- For a current $I = 3A$, we measured $\mu_B = 9.162 \times 10^{-24}$, which is quite close to the CODATA value.
- For $I = 4A$, we obtained $\mu_B = 1.069 \times 10^{-23}$, which deviates significantly from the theoretical value.

The discrepancy in our measurements might be related to our data processing method. Specifically, we used a built-in measurement tool in the software 'GIMP' to determine the inner and outer diameters. This tool operates based on the discrete pixel value method.

6 Conclusion

Through this experiment, we gained a more profound understanding of the light diffraction phenomenon. We made a comparison between two types of diffraction, namely single slit diffraction and double slit diffraction. Diffraction is characterized as the bending of waves around small obstacles and the dispersion of waves beyond small apertures. On the other hand, interference occurs when two waves superimpose to create a wave of higher or lower amplitude. The pattern produced by the interference and diffraction of coherent light exhibits distinct differences between single ,double slit and circular configurations. our exploration of the Balmer lines has provided valuable insights into the spectrum of Hydrogen and Helium.Furthermore, the Zeeman effect demonstrated the influence of a magnetic field on the energy levels,leading to the splitting of spectral lines. In summary, the experiments conducted in various areas of wave optics yielded results that varied in their alignment with theoretical expectations. Some results closely matched the theoretical predictions, while others exhibited considerable deviations. These inconsistencies underscore the challenges associated with conducting experiments and obtaining accurate results, and they highlight the importance of precise measuring tools and data processing techniques. Furthermore, it can be concluded that the sensitivity and accuracy of the devices used in the experiments play a crucial role in minimizing uncertainties and achieving reliable outcomes.

7 References

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